

Project Sentinel — Vision

Version: 0.2

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Status: Active — supersedes implicit vision in README v0.1

What Sentinel Is

Project Sentinel is an autonomous world engine. It runs a living RPG universe without a human dungeon master — maintaining narrative consistency, propagating consequences, and evolving world state across indefinite time and play sessions.

The user's only interface is the narrative. The system handles everything else.

This is not a chatbot with a fantasy skin. It is an architecture problem: how do you give an LLM the ability to *change* a shared world — reliably, consistently, without corruption — while preserving the creative chaos that makes a world worth playing in?

The answer Sentinel proposes: **schema-enforced mutation over a Git-backed state layer, with a hybrid pipeline that treats hallucination as raw material rather than failure.**

The Central Bet

Most AI-driven game systems fail in one of two ways:

1. **Too loose** — the LLM invents freely, the world drifts, continuity collapses. The goblin king you befriended three sessions ago doesn't remember you. The cave you cleared is full again. The world has no memory.
2. **Too rigid** — every possible thing is pre-defined, every branch scripted, every NPC dialogue tree written. The world is consistent but inert. Nothing emerges. Players feel the rails.

Sentinel bets on a third path: **design the physics, let the content emerge.**

Schemas encode the *laws* that govern existence — NPC goals and desires, faction tensions and resources, item histories, regional geography. The DM generates narrative *from* world state. Quest experiences, NPC dialogue, environmental conditions — none of it scripted. All

of it internally consistent with what the world currently is.

This is the difference between a world that recites and a world that thinks.

Architectural Vision

v0.1 — What the README Described

The README captured a correct skeleton:

- Three agents: DM, Fact-Extractor, Lorekeeper
- A binary Fact-Extractor: accept or reject
- A synchronous canonical write path
- MCP Bridge as the schema enforcer
- Git-backed hybrid storage (JSON state + Markdown lore)

This is sound. The separation of concerns — Inference Node never touches files, Infrastructure Node is the persistence layer, MCP Bridge is the enforcer — is a design principle worth keeping. The core loop (Action → Narrative → Extract → Validate → Persist → Reload) is the right shape.

What v0.1 didn't solve: **what happens at world generation, before schemas exist for what the DM is inventing?**

v0.2 — What the Architecture Analysis Added

The research phase identified a central paradox: schema validation is both the system's strength and its vulnerability. At the moment of maximum creative freedom — early world generation — the system has the fewest schemas. Every DM invention is either validated against an existing schema or rejected.

A binary Fact-Extractor at world generation means the system rejects the most interesting things it produces.

The resolution: a three-outcome Fact-Extractor with a `defer` path, and a parallel async pipeline that processes deferred content without blocking the canonical write path or the player experience.

This is not a workaround. It is the intended behavior of the system at full maturity.

The Hybrid Context Pipeline

The mature architecture operates two parallel paths:

Synchronous Canonical Path

DM → Fact-Extractor (accept) → MCP Bridge → persist → Lorekeeper → DM

Fast. Strict. Unchanged from v0.1. This path handles content that matches existing schemas.

Asynchronous Exploratory Path

Fact-Extractor (defer) → Exploratory Harvester → Archive
Archive → Contextual Injector → Lorekeeper → DM (as labeled context)
Archive → Schema Induction → candidate schemas → Promotion → MCP Bridge → canonic

Async. Does not block gameplay. Deferred content doesn't disappear — it enters an Exploratory Archive, gets embedded and tagged, and becomes available for context injection and eventual schema promotion.

The key principle: the DM treats injected exploratory content as rumor, not fact. Canonical world state is unconditional trust. Exploratory context is clearly labeled as unvalidated. The architecture enforces this distinction structurally — canonical facts in one context block, exploratory observations in another.

Agent Roster (v0.2)

The system grows from 3 agents to 10:

Agent	Node	Role
DM Agent	Inference	Generates narrative and candidate <world_update> payloads
Fact-Extractor	Inference	Routes payloads: accept / reject / defer
Lorekeeper	Inference	Composes RAG context from canonical + compressed lore
		Scores and injects archival content into

Contextual Injector	Inference	Lorekeeper context
Contextual Veto Agent	Inference	Rejects schema-valid but contextually invalid payloads
Exploratory Harvesters	Inference	Extract structured examples from deferred narrative
Moderation Agent	Inference	Enforces conduct and 4th-wall rules pre- and at-inference
External Events Agent	Inference	Emits background world events while players are absent
Schema Induction Service	Infrastructure	Clusters archive examples, proposes candidate schemas
Promotion Service	Infrastructure	Migrates exploratory artifacts to canonical state on cadence

The MCP Bridge expands responsibilities accordingly: promotion events, moderation audit logs, exploratory record tagging in addition to existing canonical writes.

Schema Philosophy

Old Model

Design schemas for every possible thing that could exist. Enumerate dialogue trees, quest waypoints, encounter tables, obstacle definitions.

Result: Exhausting to maintain. Brittle. Finite. The world can only contain what was pre-imagined.

New Model

Design schemas for the *laws* that govern existence. Encode physics, not inventory.

NPC schema encodes: goals, desires, affiliations, knowledge. Not dialogue trees.

Quest schema encodes: goal item, location, who knows about it. Not waypoints or branches.

The DM generates dialogue from who the NPC *is* and what is *actually happening* in the world. The quest experience emerges from coherent world state. Neither was scripted.

Bootstrap Minimum

Five schema domains sufficient to plant fertile soil:

1. **Physical Geography** — regions, terrain, climate, boundaries
2. **Faction Landscape** — control, relationships, goals, tensions
3. **Item Ontology** — what exists, where, significance, history
4. **NPC Core Traits** — goals, desires, affiliations, knowledge
5. **Event Seeds** — things that happened that shaped current state

Sparse initial population. The world fills itself through play. Schema Induction discovers what new schemas the world needs through play patterns — not through upfront design.

What Changes Downstream

Architecture (ARCHITECTURE.md)

- Node responsibility split updated: Inference Node now hosts Contextual Injector, Veto, Harvesters, Moderation; Infrastructure Node hosts Archive, Schema Induction, Promotion as scheduled jobs
- Three canonical data states: `CANONICAL` , `EXPLORATORY` , `REJECTED`
- Context injection priority rules must be formally documented
- MCP Bridge scope expansion: promotion events, audit logs, exploratory tagging

Schemas (/schemas)

- Five bootstrap domain schemas needed before first world generation
- Exploratory record tagging schema needed before async path can operate
- Existing `apply_world_update.schema.json` unaffected — canonical path unchanged

Agents (/world-engine/agents)

- `fact-extractor.yaml` — update persona to describe three outcomes and deferral criteria
- `lorekeeper.yaml` — update to describe context injection priority and contamination prevention

- Five new agent YAML files needed for new agent roster

New Documentation

- `docs/HYBRID_CONTEXT_PIPELINE.md` — motivation, defer semantic, archive design, induction cadence, promotion governance, contamination prevention
- `docs/SCHEMA_BOOTSTRAP.md` — physics-not-inventory philosophy, five bootstrap domains, what deliberately not to schema, population guidelines

Core Loop Diagram (README.md)

- Must be updated to show the defer path and async pipeline alongside the synchronous canonical path
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What Does Not Change

- The Inference Node never touches files directly. This is non-negotiable.
 - The MCP Bridge is the enforcer of all world mutations.
 - Git-backed hybrid storage (JSON state + Markdown lore) is the persistence model.
 - The canonical synchronous path is fast, strict, and unchanged.
 - Tailscale mesh as the multi-node transport layer.
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Design Tensions to Watch

Contamination risk is real. The Contextual Injector feeds exploratory archive content into the Lorekeeper. If the DM builds on unvalidated content and that content propagates, hallucination compounds. The mitigation — structural separation of canonical and exploratory context, clear labeling — must be enforced architecturally, not just via prompt instructions. Prompt engineering is fragile. Context block structure is not.

Schema Induction cadence needs a triggering philosophy. Too frequent and you formalize patterns before you have enough examples. Too infrequent and the exploratory archive accumulates debt. The right trigger is probably archive depth per topic cluster, not a time-based schedule.

External Events Agent is underspecified. It runs world physics between sessions. If it contradicts in-session canonical state, coherence collapses. It needs strong schema

constraints and a read-before-write discipline against current world state before it emits anything.

Summary

Sentinel's v0.1 README described the skeleton correctly. The architecture analysis confirmed the skeleton is sound and identified what it was missing: a path for the world to teach the system what it needs to become.

The hybrid pipeline — defer, archive, induce, promote — is not complexity added on top of the system. It is the mechanism by which a sparse, planted world becomes a rich, coherent one over time. The world grows through play. That is by design.

The humans set the physics. The content emerges.